

ENAE 410 – Design Lab 3

Stability and Trim Considerations

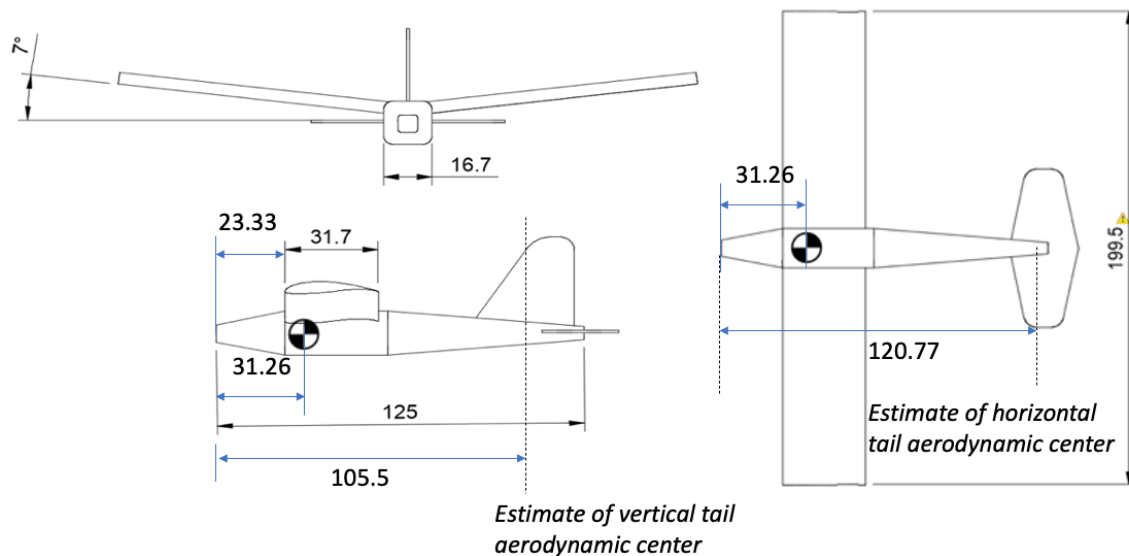
1 Objective

The core aim of this laboratory exercise is to take you through the process of verifying the static stability of the reference aircraft provided for the class. Through this hands-on activity, students will not only understand these fundamental concepts but also gain practical experience in conducting essential aviation analyses.

2 Background

In the complex world of aircraft design, stability and trim play pivotal roles, ones that were not explored in our previous analysis of an aircraft's flight envelope. When we talk about stability in aircraft, we refer to the aircraft's ability to maintain its attitude and to return to its original flight condition after being disturbed. This requires a negative lift on the horizontal tail, which, while aiding stability, simultaneously decreases the overall lift and increases the lift-induced drag. Achieving this balance in steady flight is referred to as trim. Both these elements - stability and trim, significantly influence the performance of an aircraft. A more stable aircraft requires fewer pilot interventions, which translates to lower fuel consumption and, hence, greater endurance. Trim optimization can help decrease drag, thereby extending the aircraft's range.

IMPORTANT: For this lab, you should use the suggested aircraft size you created in **Design Lab 1**. As well you may assume that the location of the center of gravity is at the $\frac{1}{4}$ chord position of the airplane. We will cover how to actually calculate a center of gravity in class.



These dimensions should be rescaled to match your "design" from Lab 1.

3 Analysis and Discussion

3.1 Calculation Procedure

3.1.1 Static Longitudinal Stability Analysis

The first step in our analysis is to assess the static stability of the aircraft, a critical factor in the aircraft's safety and performance. Here are the sub-steps:

Calculate Tail Volume Coefficients: Begin by calculating the tail volume coefficients for both the vertical and horizontal tail surfaces. The tail volume coefficient is a dimensionless number that represents the influence of the tail on the aircraft's stability. It is given by the formula:

$$V_H = \frac{l_H S_H}{S_w c}$$

where l_t is the tail arm (distance from the center of gravity to the tail aerodynamic center (assume $\frac{1}{4}$ chord)), S_t is the tail area, S_w is the wing area, and c is the mean aerodynamic chord of the wing.

$$V_v = \frac{l_v S_v}{S_w b}$$

The tail volume coefficients should be within an acceptable range. For most aircraft, the horizontal tail volume coefficient (V_H) ranges between 0.3 and 0.6, and the vertical tail volume coefficient (V_v) between 0.02 and 0.05. Is the reference aircraft in this range?

Compute the neutral point (aerodynamic center) of the airplane: Recall that the aerodynamic center (or neutral point) is the location about which the moment is constant with lift coefficient (or angle of attack). To find the aerodynamic center location of the whole aircraft, we can use the following approximation:

$$x_{ac} = x_{acwb} + \frac{V_H a_t}{a} c$$

Where x_{acwb} is the aerodynamic center of the wing-body alone (you may assume it is located at the $\frac{1}{4}$ chord position of the main wing), a_t is the lift curve slope of the horizontal tail, a is the lift curve slope of the main wing, and V_H is the tail volume of the horizontal tail. You should compute the lift curve slope of the horizontal tail using its aspect ratio and assuming a 2D lift curve slope of 2π . Enough information has been provided in the previous lab manuals to perform this calculation.

Alternatively, you can change the point about which you measure the moment in OpenVSP. When you find the location that makes the slope of C_m vs C_L zero (mostly) you have approximately found your neutral point.

Calculate Static Margin: Next, calculate the static margin, which is the distance from the center of gravity (CG) to the neutral point (NP) of the aircraft, expressed as a percentage of the mean aerodynamic chord (MAC).

$$\text{Static Margin} = (x_{ac} - x_{cg})/MAC * 100\%$$

A positive static margin indicates that the CoG is ahead of the aerodynamic center, which is generally desired for a stable aircraft. For a conventional aircraft, a typical value is 10%.

Longitudinal Stability Requirements: Finally, verify the requirements for static stability. You should use OpenVSP to calculate the moment coefficient (c_m) over a range of angles of attack. The moments described here should be about the aircraft's center of gravity. For simplicity, in this exercise, you may continue to assume that the CoG is located at 25% of the wing chord. For the aircraft to be statically stable, the following conditions are **required**:

- ☐ c_m at $\alpha = 0$ is > 0 – or alternatively: c_m at $C_L = 0$ is > 0
- ☐ $\frac{dc_m}{d\alpha}$ must be < 0

The first requirement guarantees that there is a point where the pitching moment is at zero which has a positive angle of attack (i.e. can lift). The second requirement is that the slope of c_m with respect to alpha ($\frac{dc_m}{d\alpha}$) must be negative. This means that as the angle of attack increases/pitches up, the aircraft restoring moment increases - thereby maintaining stability.

As you can see from these processes, if you do not have sufficient static margin, and if you do not meet the above requirements, you may move your center of gravity forward, or move your aerodynamic center aft by increasing V_H .

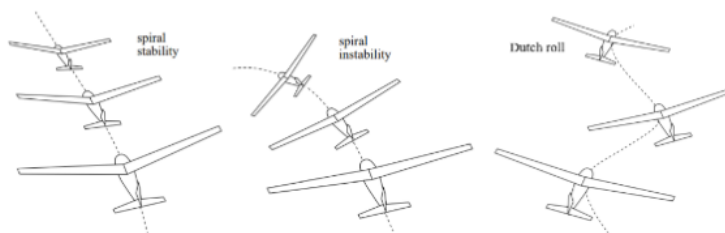
3.1.2 Spiral Stability Requirements

The spiral stability of an aircraft refers to the way in which the aircraft responds to roll. A spirally unstable aircraft will have a tendency to roll unpredictably or periodically (known as Dutch roll). This is dangerous because when the wings roll to a vertical position, all lift is lost, and the plane will fall from the sky. A spirally stable aircraft will have a tendency to return to upright flight in spite of pilot input or crosswind.

The spiral stability of an aircraft can be determined using the spiral stability parameter:

$$B = \frac{l_v \Gamma}{b C_L}$$

Where l_v is the moment arm length of the vertical tail, Γ is the dihedral angle, b is the wingspan, and C_L is the wing lift coefficient. Typically, a C_L related to a cruise / steady level flight condition is chosen. It is desirable to have a value of $B > 5$ to ensure spiral stability. Passive spiral stability is not always a firm requirement, as pilot experience and sufficient aileron control can often be sufficient.



3.1.3 Calculating Trim Drag

The final step in this lab is to compute the trim drag and investigate the impact of trim on performance. This is done by determining the negative lift needed on the horizontal tail for stability and then converting this negative lift into induced drag using the appropriate drag equations. At each angle of attack:

- Compute the speed required for flight based on:

$$V = \left[\frac{2mg}{\rho S_w \left(C_L + \frac{c C_M}{l_H \cos \alpha} \right)} \right]^{0.5}$$

- The lift force on the tailplane is then given by:

$$L_{tail,trim} = \frac{0.5 \rho V^2 C_M S_w c}{l_H \cos \alpha}$$

$$C_{L,tail,trim} = \frac{C_M S_w c}{l_H S_H \cos \alpha}$$

- Use this lift force to compute induced drag on the tailplane.

$$D_{i,tail} = \frac{L_{tail,trim}^2}{0.5 \rho V^2 \pi e b_H^2}$$

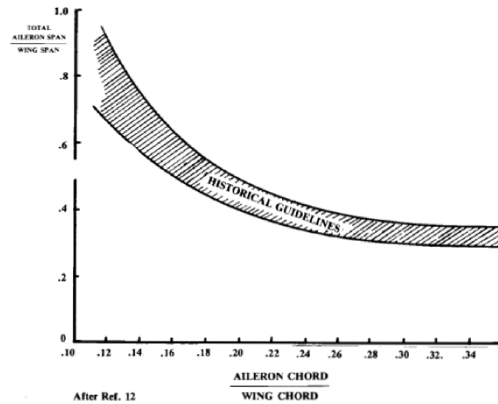
Where e is the Oswald efficiency factor (you may assume 0.8 for this lab), and b_H is the span of the horizontal stabilizer.

- Compute a new required thrust curve for your airplane and compare it to the curve without accounting for trim, in this case did it make much of a difference? What might change in the design that could cause the trim to much more important? *You may notice that this is a fairly substantial simplification. In reality, we actuate the elevator to achieve the required lift on the tail. This effectively changes the airfoil of the tail. This also happens at different angles of attack! If we wanted to be very rigorous a more complicated analysis would need be performed where we actually evaluate the required elevator deflection at each angle and the associated parasitic and induced drag. Additionally, the presence of stabilizers in the downwash or upwash of the main wing changes things greatly. In this simplified analysis, we are simply adding the induced drag of the required lift on the tail to the drag we computed before (either from drag build up or OpenVSP) and neglecting changes in parasitic drag.*

3.1.4 Size control surfaces (Rudder, Aileron, Elevator)

Briefly go through the process of sizing some control surfaces for the reference airplane. Using the following rules of thumb, describe a feasible design of the rudder, ailerons, and elevator on the reference aircraft. What factors might you consider when placing these surfaces?

Ailerons: Typically the ailerons span is 50% to 90% of the total wing span. The chord length of ailerons is usually between 15% and 25% of the wing chord. A useful plot has been provided by Raymer [1]:



Rudder and Elevator: The elevator span often extends the entire length of the total available horizontal tail, and the chord takes up 25% to 50% of the tail.

3.2 Discussion

Following the calculation procedure, a discussion is required to ensure a comprehensive understanding of the activity. In particular, please discuss the findings from the static stability analysis: would the reference aircraft be stable? Are there any improvements that could be made? Are these improvements sacrificing some stability in the name of other performance, or maybe vice-versa?

4 Summary and Expected Outcomes

By the end of this laboratory exercise, you should have a clear understanding of how stability and trim affect an aircraft's performance. You should be capable of sizing control surfaces using volume coefficients, understanding the significance of negative lift required to achieve zero net moments, and verifying static stability through analysis of static margin, pitching moment coefficient, and finding the aerodynamic center. You should be able to see how the size of the horizontal stabilizer and its location relative to the center of gravity impact the aerodynamic center and, by extension, static margin and stability. Moreover, you should be able to connect these elements to the practical aspects of aircraft design, particularly the impact on the lift, drag, and mass of the aircraft.

5 References and Supporting Materials

[1] OpenVSP official website: <http://www.openvsp.org/>

[2] OpenVSP User Manual: http://www.openvsp.org/wiki/doku.php?id=user_manual

[3] Drag buildup models: <https://openvsp.org/wiki/doku.php?id=parasitedrag>

[4] Raymer, D. (2012). Aircraft design: a conceptual approach. American Institute of Aeronautics and Astronautics, Inc..